

STATISTICS FOR DATA ANALYSIS - DAY 06 NOTES

Master Class Notes: Complete IQR Breakdown (Odd vs Even Cases) & Shape of Data (Symmetry vs Skewness)

1. Interquartile Range (IQR) Deep-Dive / IQR का विस्तृत विश्लेषण

English: IQR (Interquartile Range) represents the middle 50% of our dataset. It is completely unaffected by extreme values or outliers because it focuses purely on the central boundary of the data.

Hindi: IQR (Interquartile Range) hamare data ke beech ka 50% hissa hota hai. Ye extreme values (outliers) se bilkul mutasir nahi hota kyunki iska poora focus data ki central boundary par hota hai.

$$IQR = Q3 - Q1$$

English: To find IQR, we divide our data into 4 equal quarters. However, the exact mathematical calculation for finding positions changes slightly based on whether the total number of elements is Odd or Even.

Hindi: IQR nikalne ke liye hum data ko 4 barabar hisson (Quarters) mein baant-te hain. Lekin Total Elements ka count Odd (bisaam) hai ya Even (saam), iske mutabik positions nikalne ka tareeqa halka sa badal jata hai.

2. IQR Case 1: Odd Number of Elements (Total = 9 Customers) / विषम संख्या वाले एलिमेंट्स

English: Let's assume 9 customers visited your jewelry showroom and purchased the following gold weights (arranged in sorted format):

Hindi: Maan lijiye aapke showroom par 9 customers aaye aur unhone ye gold purchases ki (Sorted format mein):

Sorted Data Set: [10, 12, 15, 18, 20, 22, 25, 30, 35]

Manual Calculation Strategy / मैनुअल गणना की रणनीति:

English: When total elements (n) are Odd, finding the Median gives us a single clear center point. For sub-halves, we will exclude this Median to calculate exact quartile positions.

Hindi: Jab total elements (n) Odd hote hain, toh Median nikalte waqt ek single clear center value milti hai. Sub-halves nikalne ke liye hum is Median ko exclude karke exact position pata karenge.

- **Median (Q2) Position:** $Position = (9 + 1) / 2 = 5th\ term$. Our Median (Q2) is 20.
- **Lower Half (for Q1):** Elements before the median: [10, 12, 15, 18]. Here, sub-count $n = 4$.
Q1 position = $(4 + 1) / 2 = 2.5th\ term$. Average of 2nd & 3rd term: $(12 + 15) / 2 = 13.5$.
- **Upper Half (for Q3):** Elements after the median: [22, 25, 30, 35]. Here, sub-count $n = 4$.
Q3 position = $2.5th\ term$ of this upper half. Average of 2nd & 3rd term: $(25 + 30) / 2 = 27.5$.

$$IQR = Q3 - Q1 = 27.5 - 13.5 = 14.0\ Grams$$

3. IQR Case 2: Even Number of Elements (Total = 10 Customers) / सम संख्या वाले एलिमेंट्स

English: Now, let's see how calculations change when we have an even number of customer entries:

Hindi: Agar hamare paas data mein 10 customers ka weight record ho, toh calculations kaise badlengi dekhiye:

Sorted Data Set: [10, 12, 15, 18, 20, 22, 25, 30, 35, 40]

Manual Calculation Strategy / मैनुअल गणना की रणनीति:

- **Median (Q2) Position:** $Position = (10 + 1) / 2 = 5.5th \text{ term}$. Average of 5th (20) and 6th (22) terms: $(20 + 22) / 2 = 21$.
- **Lower Half (for Q1):** Since there is no single middle element, the lower half contains the first 5 elements: [10, 12, 15, 18, 20]. Here, $n = 5$ (Odd).
 $Q1 \text{ Position} = (5 + 1) / 2 = 3rd \text{ term}$. The 3rd term of this lower half is **15**.
- **Upper Half (for Q3):** The remaining last 5 elements: [22, 25, 30, 35, 40]. Here, $n = 5$ (Odd).
 $Q3 \text{ Position} = (5 + 1) / 2 = 3rd \text{ term}$. The 3rd term of this upper half is **30**.

$$IQR = Q3 - Q1 = 30 - 15 = 15.0 \text{ Grams}$$

Business Logic Note / व्यावसायिक तर्क:

English: In even datasets, when we split lower and upper halves, if the individual sub-halves contain an odd count of items, we get direct middle integer values without needing fractional averages.

Hindi: Even data mein jab lower aur upper half separate hote hain, toh agar elements ka individual count odd nikalta hai (jaise yahan 5-5 nikala), toh hamein seedhe center number mil jata hai bina kisi fractional average ke.

4. Shape of Data: Symmetry (Symmetrical Distribution) / सममितीय वितरण

English: Symmetrical distribution occurs when a dataset looks balanced on both sides of the center point. If you place a mirror at the middle, the left and right halves mirror each other completely, forming a classic bell-shaped curve.

Hindi: Symmetrical distribution ka matlab hai ki jab hum data ko graph par utarte hain, toh wo ekdum balanced dikhta hai. Agar center mein aaina (mirror) rakh dein, toh dono sides ek jaise dikhte hain. Iska curve perfectly Bell-Shaped hota hai.

Golden Rule / मुख्य नियम:

$$\text{Mean} = \text{Median} = \text{Mode}$$

Gold Coin Machine Example / गोल्ड कॉइन मशीन उदाहरण:

English: A coin stamping machine creates 5 gold coins with a target of 15 grams. Data: [14, 15, 15, 15, 16].

Hindi: Machine ne dinbhar mein 5 coins banaye jinka target weight 15 Grams hai. Data hai: [14, 15, 15, 15, 16].

- **Mean:** $(14 + 15 + 15 + 15 + 16) / 5 = 75 / 5 = 15$
- **Median:** Sorted list central term = 15
- **Mode:** Most frequent term (15 appeared 3 times) = 15

English: All three metrics match perfectly. The distribution on the left is identical to the right, showing zero distortion or skewness.

Hindi: Yahan teeno metrics bilkul barabar hain. Center (15) ke piche jitna bikhraav hai (14), aage bhi bilkul utna hi bikhraav hai (16). Isliye data me koi jhukav ya bias nahi hai.

5. Shape of Data: Positive Skewness (Right-Skewed) / धनात्मक विषमता

English: Positive skewness occurs when the major peak of the data is on the left side (smaller values), but a few extremely large values pull a long tail towards the right side.

Hindi: Positive Skewness tab hoti hai jab graph ka bada pahad left side (choti values) mein hota hai, lekin bade heavy outliers ki wajah se ek lambi ponch (tail) right side ki taraf khinch jati hai.

Golden Rule / मुख्य नियम:

$$\text{Mean} > \text{Median} > \text{Mode}$$

Showroom VIP Customer Example / वीआईपी ग्राहक उदाहरण:

English: Regular customers purchase small daily rings, but suddenly a single VIP client walks in and buys an exceptionally heavy custom bullion block.

Hindi: Roz normal log aate hain jo choti rings lete hain, par ek din kisi VIP ne achanak bada heavy item kharid liya.

Data Set: [10, 12, 12, 15, 101] — (101 Grams is a heavy outlier)

- **Mode:** 12 (Appears most frequently)
- **Median:** Center value = 12
- **Mean:** $(10 + 12 + 12 + 15 + 101) / 5 = 150 / 5 = 30$

Business Impact / व्यावसायिक प्रभाव: That single high-value bulk buyer shifted our average from a routine 12 grams straight up to 30 grams, pulling the distribution tail rightwards. / Us ek heavy customer ne routine average ko 12 se seedhe 30 par khinch liya. Ponch right side bhaag gayi.

6. Shape of Data: Negative Skewness (Left-Skewed) / ऋणात्मक विषमता

English: Negative skewness happens when most data points cluster on the higher side (right), but a few extremely small abnormal values pull the graph's tail towards the far left.

Hindi: Negative Skewness tab hoti hai jab zyadatar data right side (badi values) mein hota hai, par kuch bohut choti extreme values ki wajah se graph ki ponch (tail) left side ki taraf khinch jati hai.

Golden Rule / मुख्य नियम:

$$\text{Mean} < \text{Median} < \text{Mode}$$

Hit Product Strike Day Example / हिट प्रोडक्ट और हड़ताल का उदाहरण:

English: A top-selling item performs beautifully every day, but a sudden market strike or extreme flood forces a complete lockdown day with zero sales.

Hindi: Ek super hit jewelry set roz badiya bikta hai, par kisi ek din strike ya bhari baarish se dukan band rahi aur zero sale hui.

Data Set: [0, 50, 52, 55, 55] — (0 sets represents a severe low outlier)

- **Mode:** 55 (Most repeated entry)
- **Median:** Central entry = 52
- **Mean:** $(0 + 50 + 52 + 55 + 55) / 5 = 212 / 5 = 42.4$

Business Impact / व्यावसायिक प्रभाव: The single zero-sales day heavily dragged down the recorded average performance to 42.4, masking the true routine store potential. / Us ek kharab din (0 sale) ne poore hit product ke average performance ko gira kar 42.4 kar diya, jabki routine sale 52-55 chal rahi thi.

7. Summary Matrix for Quick Revision / त्वरित पुनरीक्षण तालिका

Distribution Type	Tail Position (पूंछ)	Mathematical Rule	Business Context Insight
Symmetrical	No Tail (Balanced)	Mean = Median = Mode	Predictable, standard & stable operations.
Positive Skewness	Right Side (Seedhe Haath)	Mean > Median > Mode	Presence of rare VIP or massive bulk orders.
Negative Skewness	Left Side (Ulte Haath)	Mean < Median < Mode	System drops, operational shutdowns, or dry days.

Day 06 Master Pro-Tip / मास्टर प्रो-टिप: In data analytics, whenever someone highlights that a process is skewed, do not rely blindly on the Mean! Look closely at the Median and IQR to understand the true underlying reality. / Data analytics mein jab bhi koi aapko bolta hai ki dhandha "Skewed" hai, toh samajh lena average par aankh band karke vishwas nahi karna hai! Wahan sachai faisla humesha Median aur IQR dekh kar hi liya jata hai.